

Summary: Artificial Intelligence Policy Worldwide

1. Overview and Core Regulatory Trends

As artificial intelligence (AI) increasingly integrates into society, documenting emergent socioeconomic and physical harms has prompted global policymakers to move toward AI-specific legislation. This paper provides a comparative study of the evolving policy landscapes across eight key jurisdictions: the EU, the UK, China, South Korea, Singapore, Brazil, Russia, and the US.

Two primary regulatory styles have emerged globally:

- **Horizontal Approaches:** Sector-agnostic laws applying centralized, overarching rules across multiple use cases (typically using a risk-based tiering system).
- **Vertical Approaches:** Sector- or technology-specific rules targeting specialized domains like human resources (HR), autonomous vehicles, or generative AI.

2. Jurisdictional Analysis

A. The European Union (EU)

- **Primary Instrument:** The Union-level **EU AI Act** (entered into force August 1, 2024; staggered enforcement starting 2025–2026).
- **Regulatory Mechanism:** A strict risk-based framework.
 - *Prohibited Systems:* Systems that present unacceptable risks (e.g., subliminal manipulation, emotion recognition in workplaces, untargeted facial scraping) are banned.
 - *High-Risk Systems:* Strict pre-market obligations (risk management, data governance, human oversight, logging) apply to employment, biometrics, critical infrastructure, and education.
 - *General-Purpose AI (GPAI):* Mandatory technical documentation, copyright policies, and training data summaries.
- **Penalties:** Drastic administrative fines up to €35 million or 7% of total worldwide annual turnover.
- **Enforcement State:** Driven significantly by existing GDPR data protection compliance mechanisms, with Italy and Spain issuing the highest volume of fines. Member states simultaneously pursue localized laws, primarily regarding automated transport and employment.

B. The United Kingdom (UK)

- **Regulatory Approach:** Historically pursued a decentralized, non-statutory, "pro-innovation" approach relying on existing regulators and core principles (Safety, Transparency, Fairness, Accountability, Contestability).
- **Legislative Shift:** Transitioning toward concrete legal frameworks to expand regulatory capabilities. Key bills introduced include the *Artificial Intelligence (Regulation) Bill* (proposing a central AI Authority and regulatory sandboxes) and the *Public Authority Algorithmic Systems Bill*.
- **Passed Legislation:** The **Automated Vehicles Act** (Autumn 2024) is the UK's first passed AI-specific law, focusing on safety and licensing.
- **Enforcement:** Led predominantly by the Information Commissioner's Office (ICO) targeting unlawful data profiling and processing violations under the UK GDPR.

C. China

- **Regulatory Approach:** Highly centralized, focusing tightly on vertical restrictions, deepfakes, online content moderation, and algorithmic influence on public discourse.
- **Key Enactments:** * *Algorithm Recommendation Regulations (2022)*: Controls mobile application algorithmic recommendations, bans price discrimination, and protects user selection choices.
 - *Deep Synthesis and Generative AI Measures (2023)*: Restricts generative models, requiring content verification, compliance with core socialist values, and non-discrimination.
 - *Synthetic Content ID (2025)*: Mandates implicit and explicit watermarking on all AI-generated content.
- **Industry Promotion:** Devolved to localized levels (e.g., Shanghai and Shenzhen), using localized sandboxes and granting safe harbors for minor infractions to protect tech growth.

D. South Korea

- **Primary Instrument:** The **AI Basic Act** (passed December 2024, taking effect January 2026).
- **Regulatory Mechanism:** Represents a layered, risk-based posture but remains intentionally more innovation-friendly than the EU. It avoids rigid pre-market conformity barriers, opting instead for a voluntary certification-based trust mechanism and post hoc enforcement oversight.
- **Focal Points:** Establishes heightened transparency and notification constraints for "high-impact" sectors (energy, water, healthcare, transportation) and generative AI.
- **Penalties:** Moderate deterrence; penalties are legally capped at 30 million won (approx. €21,500), missing an explicit civil liability framework.

E. Singapore

- **Regulatory Approach:** Strongly resilient against rigid horizontal legislation, prioritizing agile, principles-based, voluntary frameworks.
- **Key Frameworks:** The *Model AI Governance Framework* (updated for Generative AI in 2024) and **AI Verify**—a dedicated technological open-source toolkit allowing corporate entities to test and validate system performance.
- **Agentic AI Leadership:** Globally unique in releasing targeted guidelines (**Agentic AI Primer** in 2025) and testing mechanisms via the *Global AI Assurance Pilot* to handle advanced autonomous software agents.
- **Liability Clearances:** Offers explicit safe-harbor guidelines protecting developers from generative output liabilities if reasonable safeguards, labeling, and redress portals are structurally present.

F. Brazil

- **Primary Instrument:** A successive pipeline of legislative proposals (Bill 5051/2019, Bill 21/2020, Bill 872/21) culminating in **Bill No. 2338/2023**.
- **Regulatory Mechanism:** Deeply mirrors the horizontal, risk-based layout of the EU AI Act. It explicitly sets down strict liability for damages, institutes mandatory public impact assessments for high-risk applications (including credit score filtering, employment, and healthcare), and prohibits unacceptable applications like social scoring.
- **Penalties:** Fines cap at R\$50 million (approx. €8.1 million) or up to 2% of annual revenue.

G. Russia

- **Regulatory Approach:** Minimal focus on comprehensive constraints, centering strategy strictly around regulatory sandboxes.
- **Legislation:** Utilizes Experimental Legal Regimes (Law 123-FZ in Moscow; expanded nationally via Law 258-FZ) enabling companies to test digital innovations across healthcare and transport with temporary regulatory exemptions. A 2024 amendment (169-FZ) introduced mandatory civil liability insurance for sandbox participants.

H. The United States (US)

- **Regulatory Approach:** Highly fragmented, decentralized legislative environment across federal, state, and local levels, leading to the highest volume of proposed bills worldwide.
- **Federal Strategy:** Absence of unified federal legislation due to intense corporate lobbying ("Silicon Valley Effect"). Cohesion relies on White House non-legislative

actions, primarily **Executive Order 14110**, alongside federal agency enforcement policy statements (FTC, EEOC, CFPB, SEC) using pre-existing consumer protection and civil rights statutes.

- **State Activity:** State-level acts have stepped in to enforce horizontal constraints (e.g., Colorado’s SB205 preventing algorithmic discrimination) and vertical restrictions (e.g., New York City’s Local Law 144 requiring independent bias audits for automated hiring tools).

3. Comparative Matrix of Key Requirements

The paper identifies key thematic focus areas across use cases globally:

Target Domain	Typical Global Regulatory Requirements	Representative Jurisdictions
Human Resources (HR Tech)	Independent bias audits, user notifications, anti-discrimination safeguards, right to human review.	US (NYC), EU, Spain, Italy, Brazil, UK
Autonomous Vehicles	Operational permits, special licensing, safety documentation, mandatory liability insurance.	EU (France, Germany, Netherlands), UK, US
Generative AI	Copyright registration for training data, implicit/explicit watermarking, systemic risk auditing.	China, EU, US

4. Key Takeaways for Legal and Policy Analysis

1. **The Overlap of Pre-existing Frameworks:** True systemic deterrence currently stems from pre-existing legal regimes - such as the GDPR in Europe or FTC financial/deceptive practice regulations in the US -rather than newborn AI statutes, which often carry small or undefined penalties.

2. **Unusual Legal Artifacts ("Regulating *With* AI"):** The paper documents a nascent trend where legislation itself is being natively generated by large language models. Costa Rica's AI Regulation Bill and Massachusetts Bill S.31 were both drafted directly with the assistance of ChatGPT, presenting novel questions regarding statutory accountability and legal drafting standards.
3. **The Core Obstacle to International Harmonization:** While multilateral forums (OECD, G7, UN) have created overlapping conceptual ethical pillars, true harmonization is hindered because states cannot agree on the clear boundaries of what constitutes "high-risk," nor do they maintain matching structures for enforcement or ex ante checking protocols.